Gravitation $w_{1+\infty}$ Symmetry

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Abstract

Extremely rough notes on gravitation $w_{1+\infty}$ symmetry. Maybe these will turn into something coherent, maybe not!

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1 $w_{1+\infty}$ and the Celestial Sphere

Scratch notes following [**?**].

1.1 Gravity

Denote by $G^+_\Delta(z,\bar{z})$ the celestial dual to a positive-helicity[1] bulk graviton. The authors of [**?**] find by analyzing bulk collinear limits that two such operators have OPE

$$G^+_{\Delta_1}(z_1,\bar{z}_1)G^+_{\Delta_2}(z_2,\bar{z}_2) = -\frac{\kappa}{2}\frac{1}{z_{12}}\sum_{n=0}^{\infty} B(\Delta_1-1+n,\Delta_2-1)\frac{\bar{z}_{12}^{n+1}}{n!}\bar{\partial}^n G^+_{\Delta_1+\Delta_2}(z_2,\bar{z}_2) + \mathcal{O}\left(z_{12}^0\right). \tag{1}$$

Here B is the Beta function

$$B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)} \tag{2}$$

holomorphic poles[2]. These operators have conformal spin $s=+2$ and scaling weights Δ, hence conformal weights

$$(h,\bar{h}) = \left(\frac{\Delta+s}{2},\frac{\Delta-s}{2}\right) = \left(\frac{\Delta}{2}+1,\frac{\Delta}{2}-1\right). \tag{3}$$

These operators are, by definition, the result of Mellin-transformating a bulk scattering amplitude of gravitons, necessarily in an asymptotically flat spacetime, to produce a boundary correlation function.

$$\langle G^\pm_\Delta(z,\bar{z})\cdots\rangle \equiv \int_0^\infty \frac{d\omega}{\omega}\omega^\Delta \cdots \mathcal{A}\left(\omega\hat{p}(z,\bar{z}),\cdots\right). \tag{4}$$

From the well-known property of Mellin transformations

$$\int_0^\infty \frac{d\omega}{\omega}\omega^\Delta\left(\lim_{\omega\to 0}\omega f(w)\right) = \lim_{\Delta\to 1}(\Delta-1)\int_0^\infty \frac{d\omega}{\omega}\omega^\Delta \omega f(w) \tag{5}$$

[1] TODO: Comment on choice to restrict to positive-helicity sector.
[2] TODO: Comment more on this choice.

we see that a bulk amplitude involving a soft graviton

$$\lim_{\omega\to 0} \omega \mathcal{A}\left(\omega \hat{p}(z,\bar{z}), \cdots\right) \tag{6}$$

corresponds on the boundary to taking the limit

$$\lim_{\Delta\to 1} (\Delta - 1) G^{+}_{\Delta}(z,\bar{z}). \tag{7}$$

This nicely agrees with (1), which we see has an order-one pole at $\Delta_i = 1$, rendering this limit finite. In fact we see it has a tower of order-one poles at $\Delta_i = 1, 0, -1, \ldots$. We therefore define more generally

$$H^k(z,\bar{z}) \equiv \lim_{\varepsilon\to 0} \varepsilon G^{+}_{k+\varepsilon}(z,\bar{z}),\ k \in \mathbb{Z}. \tag{8}$$

We will shortly find that these live in a shortened representation, so that only a subset of k values give a nonzero result.

These objects turn out to generate an interesting algebra. We first expand in the antiholomorphic coordinate

$$H^k(z,\bar{z}) = \sum_n \frac{H^k_n(z)}{\bar{z}^{n+\frac{k-2}{2}}} \tag{9}$$

where the shift in the powers of $\bar{z}$ is required by H^k having a nonzero antiholomorphic conformal weight. Again, we will find that the relevant values of n are limited. We now interpret the objects $H^k_n(z)$ as holomorphic currents on the boundary.